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Table of Contents

1. About Course
2. About Me
3. Course Outcomes
4. Syllabus
5. Book Recommended
6. Suggested Practicals
7. Marking Scheme

About Course

- **Course Name:** DISTRIBUTED COMPUTING
- **Course Code:** CS332
- **Course Instructors:** Dr. Naveen Kumar and Dr. Bhavesh N. Gohil
- **Number of Lectures per Week:** 3
- **Number of Lab per Week:** 1

About Me

- **Naveen Kumar** (Assistant Professor)
- Department of Computer Science and Engineering
- **Contact Information**
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- **Research Area**
  - Enforcing Security in Named Data Networking
  - Fake News Detection
  - Enforcing Privacy in Medical Image Based Diagnostics
  - Optimization: Task Scheduling, Clustering, Controller Placement

Course Outcomes

- At the end of this course, students will be able to:
  - Understand core concepts of distributed systems and design challenges
  - Design distributed algorithms for primitives such as Mutual exclusion, Deadlock detection, Agreement protocols
  - Analyze system faults and apply fault-tolerance techniques
  - Understand election algorithms, time synchronization, and naming services
  - Design distributed programs under performance and scalability constraints

Syllabus

INTRODUCTION TO DISTRIBUTED SYSTEMS	**(06 Hours)**
Review of Networking Protocols, Point to Point Communication, Operating Systems, Concurrent Programming, Characteristics and Properties of Distributed Systems, Goals of Distributed Systems,Multiprocessor and Multicomputer Systems, Distributed Operating Systems, Network Operating Systems, Middleware Concept, The Client-Server Model, Design Approaches-Kernel Based-Virtual Machine Based, Application Layering.	
COMMUNICATIONIN DISTRIBUTED SYSTEMS	**(04 Hours)**
Layered Protocols, Message Passing-Remote Procedure Calls-Remote Object Invocation, Message Oriented Communication, Stream Oriented Communication, Case Studies.	
PROCESS MANAGEMENT	**(05 Hours)**
Concept of Threads, Process, Processor Allocation, Process Migration and Related Issues, SoftwareAgents, Scheduling in Distributed System, Load Balancing and Sharing Approaches, Fault Tolerance, Real Time Distributed System.	

Syllabus (Cont'd)

	SYNCHRONIZATION	**(06 Hours)**
	Clock Synchronization, Logical Clocks, Global State, Election Algorithms-The Bully algorithm-A Ring algorithm, Mutual Exclusion-A Centralized Algorithm-A Distributed Algorithm-A token ring Algorithm, Distributed Transactions.	
	CONSISTENCY AND REPLICATION	**(06 Hours)**
	Introduction to Replication, Object Replication, Replication as Scaling Technique, Data Centric Consistency Models-Strict-Linearizability and Sequential-Causal-FIFO-Weak-release-Entry, Client Centric Consistency Models-Eventual Consistency-Monotonic Reads and Writes-Read your Writes- Writes Follow Reads, Implementation Issues, Distribution Protocols-Replica Placement-UpdatePropogation-Epidemic Protocols, Consistency Protocols.	
	FAULT TOLERANCE	**(04 Hours)**
	Introduction, Failure Models, Failure Masking, Process Resilience, Agreem in Faulty Systems, Reliable Client Server communication, Group communication, Distributed Commit, Recovery.	

Syllabus (Cont'd)

DISTRIBUTED OBJECT BASED SYSTEMS	**(06 Hours)**
Introduction to Distributed Objects, Compile Time Vs Run Time Objects, Persistent and Transient Objects, Enterprise JAVA Beans, Stateful and Stateless Sessions, Global Distributed Shared Objects,Object Servers, Object Adaptors, Implementation of Object References, Static And Dynamic Remote Method Invocations, Replica Framework.	
DISTRIBUTED FILE SYSTEMS	**(04 Hours)**
Introduction, Architecture, Mechanisms for Building Distributed File Systems-Mounting-Caching- Hints-Bulk Data Transfer-Encryption, Design Issues-Naming and Name Resolution-Caches on Disk or Main Memory-Writing Policy-Cache consistency-Availability-Scalability-Semantics, Case Studies,Log Structured File Systems.	
DISTRIBUTED WEB BASED SYSTEMS	**(04 Hours)**
Architecture, Processes, Communication, Naming, Synchronization, Web Proxy Caching, Replication of Web Hosting Systems, Replication of Web Applications.	
Practicals will be based on the coverage of the above topics separately.	**(30 Hours)**

Book Recommended

4.	**Books Recommended**
1	Andrew S Tanenbaum, “Distributed systems: Principles and Paradigms”, 4th Edition, Pearson Education. Inc 2023.
2	Mukesh Singhal and Niranjan G. Shivaratri, “Advanced Concepts in Operating Systems”, TMH, McGraw-Hill, Inc. New York, USA 1994.
3	Pradeep K. Sinha, “Distributed Operating System: Concept and design”, PHI, New Delhi 2019.
4	W Richard Stevens, “Unix Network Programming: Vol 1, Networking APIS: Sockets & XTI”, Third Edition E, Pearson Education, 2003.
5	Colouris, Dollimore, Kindberg, "Distributed Systems Concepts & Design", 5th Edition, Pearson Ed. 2011.

Distributed Systems Practicals – Tools & Implementation

- **Socket Programming using Java**
  - Language: Java (JDK 8+)
  - APIs: `java.net.Socket`, `ServerSocket`, `DatagramSocket`
  - Tasks:
    - Implement TCP client–server chat application
    - Extend server to handle multiple clients using threads
    - Implement UDP-based message exchange
- **Remote Invocation using Java RMI**
  - Tools: Java RMI, `rmiregistry`
  - Tasks:
    - Define remote interface and implementation
    - Bind service to RMI registry
    - Invoke methods remotely from client

Web and P2P

- **REST-based Web Services**
  - Framework: Spring Boot (Java)
  - Tools: Maven, Postman, REST Client
  - Tasks:
    - Create REST APIs using `GET/POST/PUT/DELETE`
    - Exchange data using JSON
    - Deploy and test services locally
- **Peer-to-Peer Overlay Network (Chord)**
  - Tools: `https://github.com/ChuanXia/Chord` or `https://sourceforge.net/projects/peersim/`
  - Platform: Multiple VMs / machines
  - Tasks:
    - Start multiple Chord nodes
    - Observe node join/leave
    - Perform key lookup and routing

Cluster-Based Practicals

- **Cluster & Process Migration (Exposure)**
  - Platform: Linux + MOSIX
  - Tasks:
    - Configure MOSIX cluster
    - Run CPU-intensive jobs
    - Observe live process migration

Marking Scheme (Tentative)

- Theory (Total 100 Marks)
  - **Mid Semester Examination**: 30 Marks
  - **End Semester Examination**: 50 Marks
  - **Quiz**: 20 Marks
- Lab (Total 50 Marks)
  - **Systematic Evaluation**: 25 Marks
  - **Group Project (Implement a Research Paper On Distributed System) and Individual Project (Solve a Real Life Problem Using Distributed Computing)**: 15 + 10 Marks